Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/24/2005
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Investigation of Future Microgravity Atomic Clocks		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	04/01/1998	**End Date:**	03/31/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	5	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2905		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity.		
Task Description:	We are developing new techniques to significantly advance atomic clocks. The microgravity environment of the ISS can lead to significantly better atomic clock due to the potentially long interrogation times. RACE (Rubidium Atomic Clock Experiment) is a flight project that will use Rb atoms to avoid the large cold collision frequency shift. It will advance atomic clock science, significantly improve the classic clock tests of general relativity, and globally disseminate time and frequency. The objective of RACE is to build a microgravity-based laser-cooled atomic clock capable of better than 10-16 accuracy and a stability approaching 10-15 for an averaging time of 1 second. These will be compared against an earth-based clock to establish the clock's accuracy and also to perform tests of general relativity. Atomic clocks have many applications ranging from navigation to the synchronization of high-speed communication networks. Developing more accurate atomic clocks is important because they may enable dramatic improvements in interplanetary navigation and potential improvements to the widely used Global Positioning System GPS. The microgravity environment of the		

	ISS can allow dramatic improvements in the accuracy of atomic clocks and also provides a natural international time and frequency dissemination capability. A comparison of clocks on the ISS and clocks on the ground will lead to improvements as large as one million on precise tests of general relativity. In addition, our measurements of cold-collision cross sections show that Rb clocks can be much better than the accepted Cs standard that defines the SI second. The new techniques we have developed have advanced the capabilities of earth-based atomic clocks, future microgravity clocks, and other precision measurements in atomic physics. As a prominent example of the impact of our measurements, laser-cooled Rb clocks are currently under construction and will be the ground reference clock for GPS beginning in 2005.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on Rubidium clocks has lead to more accurate atomic clocks. As a result, a number of Rb clocks are currently under construction and these are expected to be the ground reference clock for GPS beginning in late 2005. Our work on juggling and microwave cavity designs has also changed the designs of clocks around the world.
Task Progress:	During the past year we have published a comprehensive theoretical study of microwave cavities for atomic clocks. The paper includes analytic solutions and numerical finite-element methods that facilitate efficient modeling of microwave cavities. The paper presents an improved cavity design where the phase variations are orders of magnitude smaller than those in the cavities currently in use. The design modifications to improve the cavities are simple to implement and our analysis also shows new and straightforward methods to verify the cavity's construction. We have implemented new cooling techniques in our atomic fountain which results in atomic fountain with temperatures less than 500 nanoKelvin. This will enable the measurements needed to design ultra-high stability clocks that juggle multiple balls of atoms. We have also begun to construct an experiment to make an optical clock, a clock based on ultra-stable lasers.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Articles in Peer-reviewed Journals	Li R, Gibble K. "Phase variations in microwave cavities for atomic clocks." Metrologia. 2004 Oct;41(6):376-86. https://doi.org/10.1088/0026-1394/41/6/004 , Oct-2004
Articles in Peer-reviewed Journals	Lammerzahl C, Ahlers G, Ashby N, Barmatz M, Biermann PL, Dittus H, Dohm V, Duncan R, Gibble K, Lipa J, Lockerbie N, Mulders N, Salomon C. "Experiments in Fundamental Physics scheduled and in development for the ISS." General Relativity and Gravitation. 2004 Mar;36(3):615-49. Review. http://dx.doi.org/10.1023/B:GERG.0000010734.62571.b4 , Mar-2004